INCH-POUND

MIL-C-83513/10A
28 June 1993
SUPERSEDING
MIL-C-83513/10
11 February 1992

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, PLUG, MICROMINIATURE, POLARIZED SHELL, RIGHT ANGLE, PIN CONTACTS, 2 ROW ,SOLDER TYPE, NARROW PROFILE, 9 THROUGH 37 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification Listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-83513

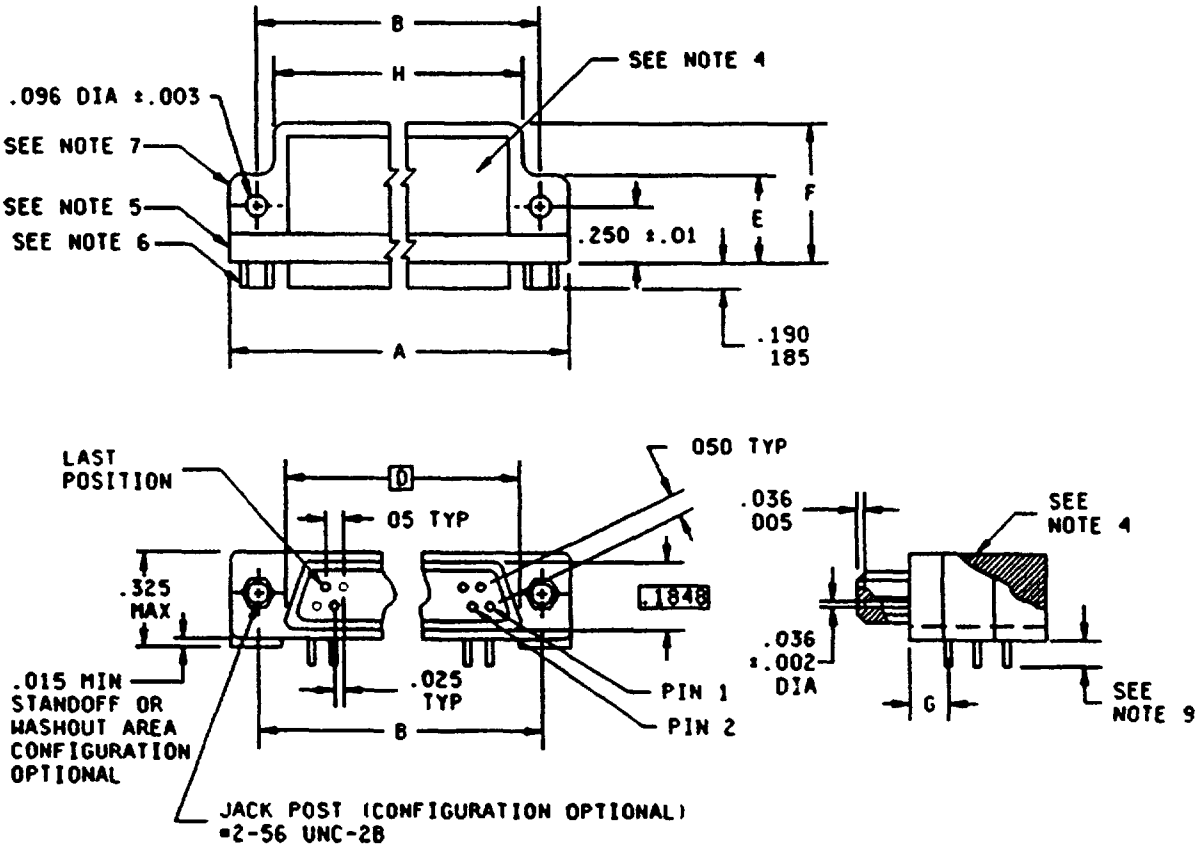
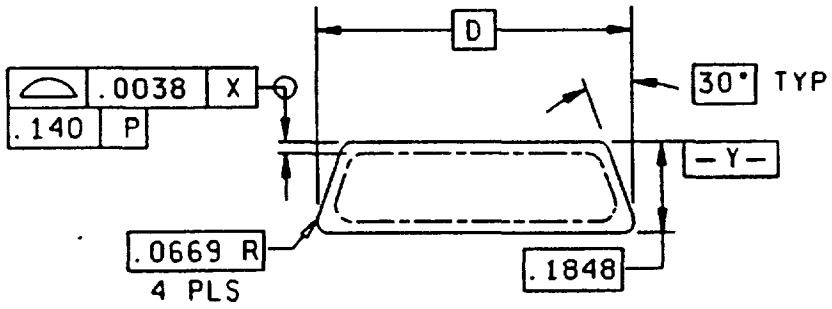


FIGURE 1 Connector, plug, 050 spacing

(A) denotes changes

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DETAIL OF INTERFACE

Inches	mm	Inches	mm
002	0 05	109	2 77
003	0 08	.140	3.56
0038	0 097	1848	4 694
010	0 25	185	4 70
015	0.38	.186	4.72
036	0 91	.190	4 83
050	1 27	250	6 35
0669	1 699	325	8 26
096	2 44		

FIGURE 1 Connector, plug, 050 spacing - Continued

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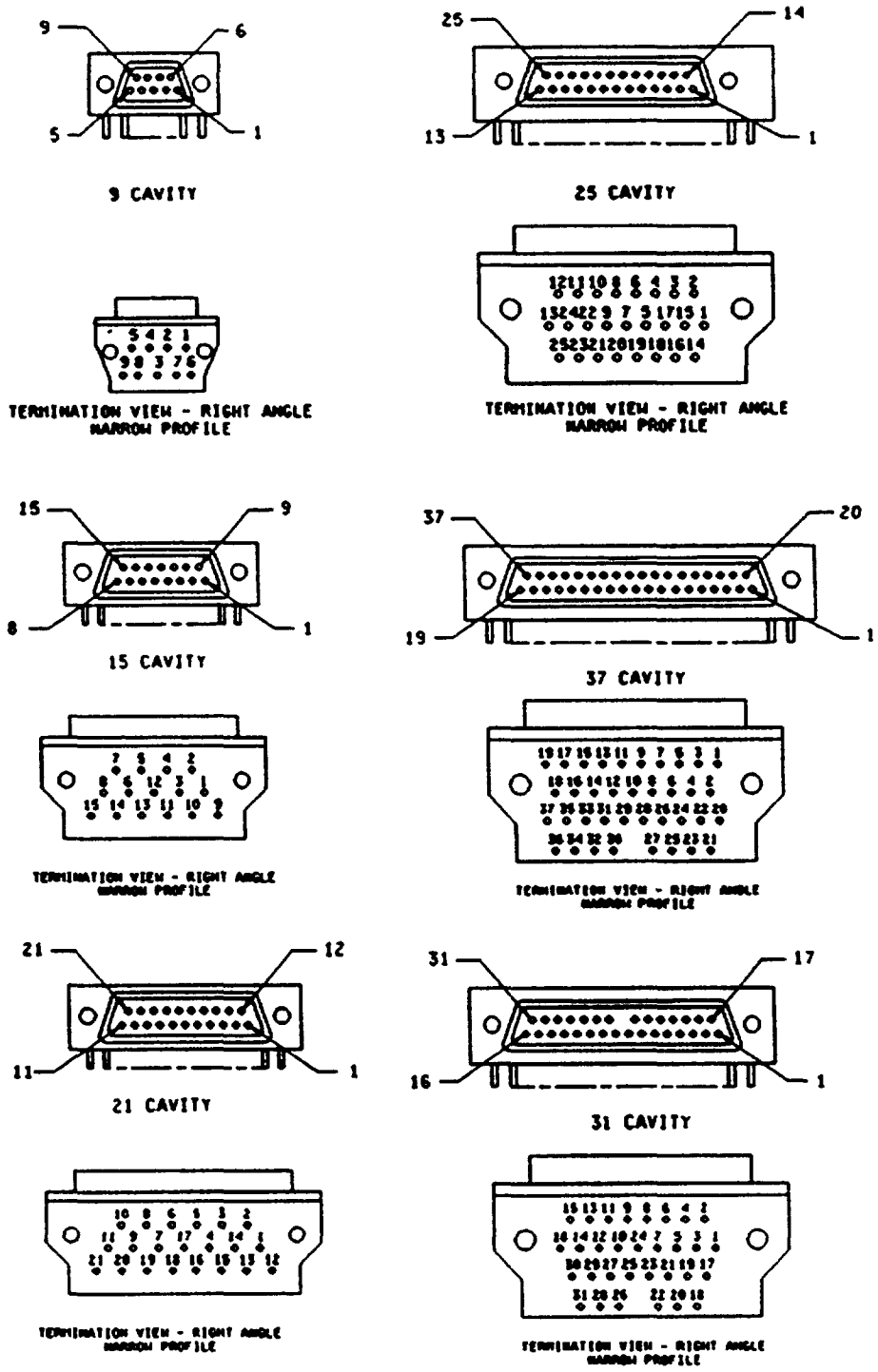
Number of contacts	A Max	B ±.005	D	E Max	F Max	G ±.01	H Max
9	.787 (19.99)	565 (14.35)	.3338 (8 48)	425 (10.80)	425 (10.80)	.230 (5.84)	787 (19 99)
15	.937 (23 80)	.715 (18.16)	4838 (12.29)	.425 (10.80)	425 (10.80)	.130 (3.30)	937 (23.80)
21	1.087 (27.61)	.865 (21.97)	.6338 (16.10)	.425 (10.80)	425 (10.80)	.130 (3.30)	1.087 (27.61)
25	1.187 (30.15)	965 (24 51)	.7338 (18 64)	.425 (10 80)	.425 (10 80)	.130 (3 30)	1 187 (30 15)
31	1.337 (33 96)	1.115 (28.32)	8838 (22 45)	.450 (11.43)	.525 (13.34)	.130 (3.30)	1.090 (27 69)
37	1.487 (37 77)	1.265 (32.13)	1.0338 (26.26)	450 (11 43)	.525 (13.34)	.130 (3.30)	1.190 (30.23)

NOTES:

1. Dimensions are in inches
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± 005 (0.13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513 Plastic molding shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519
5. Metal shell shall be of material in accordance with MIL-C-83513 for class M parts.
6. Jackpost (permanently attached), when specified Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35
7. Separately molded plastic body (if used) shall conform to the requirements of type GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
8. Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper.
9. Termination lengths available .109 (2.77 mm), .140 (3.56 mm) or 172 (4.37 mm). The tolerance shall be ± 015 (0 381 mm) for all termination lengths.

FIGURE 1 Connector, plug, 050 spacing - Continued

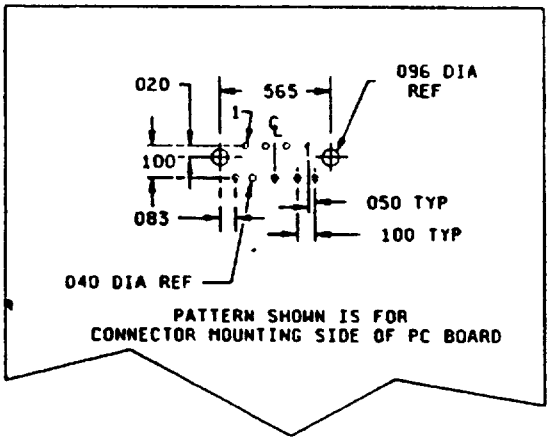
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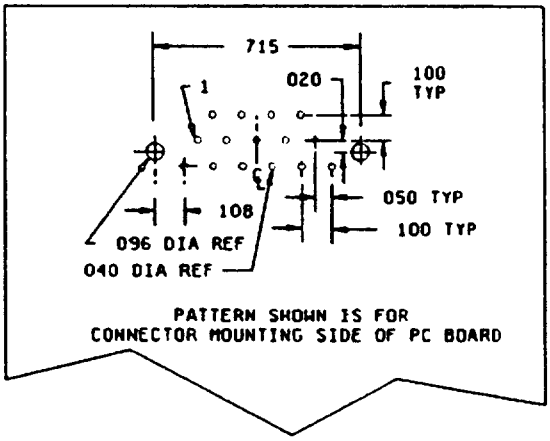
NOTE Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part

FIGURE 2 Insert arrangement, termination view - right angle - narrow profile

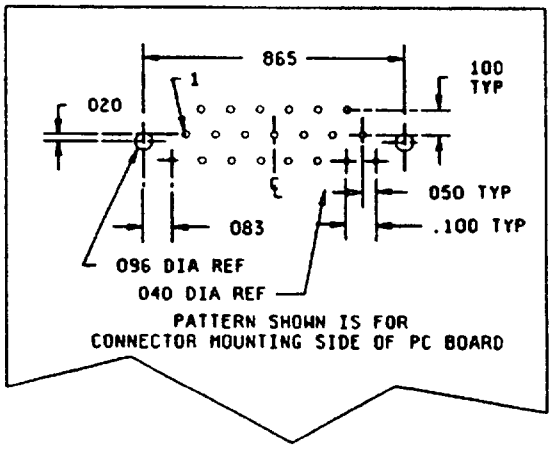
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9 CONTACT PLUG PIN CONNECTOR
RIGHT ANGLE



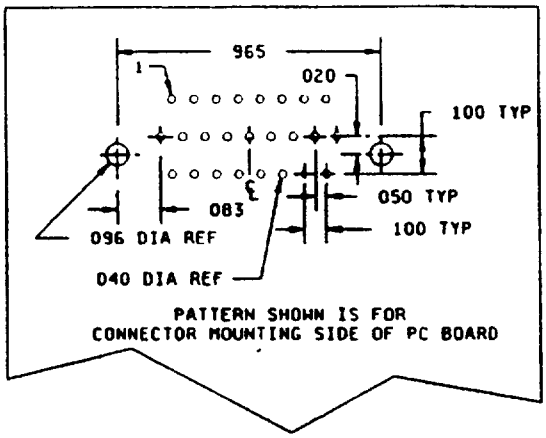
15 CONTACT PLUG PIN CONNECTOR
RIGHT ANGLE



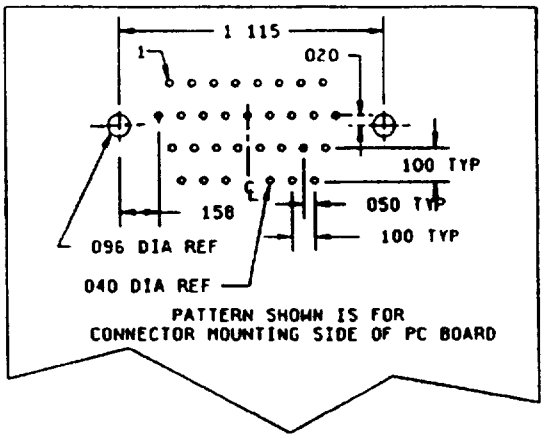
21 CONTACT PLUG PIN CONNECTOR
RIGHT ANGLE

FIGURE 3 Layout arrangement

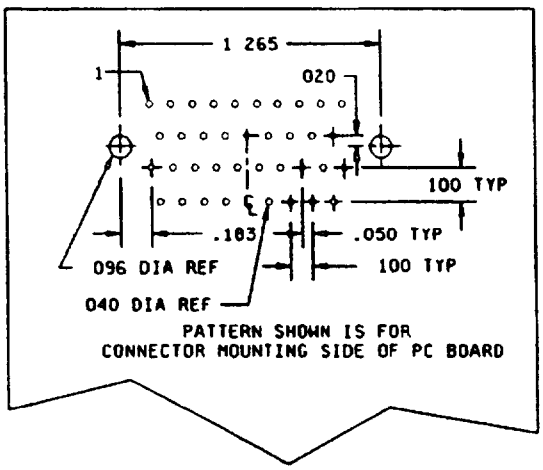
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25 CONTACT PLUG PIN CONNECTOR
RIGHT ANGLE



31 CONTACT PLUG PIN CONECTOR
RIGHT ANGLE



37 CONTACT PLUG PIN CONNECTOR
RIGHT ANGLE

Inches	mm	Inches	mm
.020	0 51	158	4 01
040	1 02	565	14 35
050	1.27	715	18 16
083	2 11	865	21.97
096	2 44	.965	24.51
100	2 54	1 115	28 32
108	2 74	1 265	32 13

NOTES

1 Dimensions are in inches
2 Metric equivalents are given for information only
3 Unless otherwise specified, tolerances are ± 005 (0 13 mm)

FIGURE 3 Layout arrangement - Continued

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REQUIREMENTS

Dimensions and configurations See figures 1 and 2

Current rating, maximum 3 amperes per contact

Materials:

Termination organization area Potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513

Shell The requirements for shell materials shall be in accordance with MIL-C-83513

Plastic body or plastic molding Shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F, or GLCP-30F, or GLCP-50 of MIL-M-24519.

Jackpost Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35

Wire termination pins Wire termination pins shall conform to QQ-W-343, no 24 AWG copper

Mating connector Shall conform to MIL-C-83513/2 and MIL-C-83513/4

Plating of termination leads. Solder dipping of termination leads will be accomplished in SN60 PB40 or SN63 PB37 in accordance with QQ-S-571

Part or Identifying Number (PIN): The PIN shall consist of the letter M, the basic number of the specification sheet, a letter for the insert arrangement, a numerical code for the termination length, and a letter code from the shell finish and hardware columns

M83513/10-	A	01	C	P
Specification sheet number	Insert arrangements (see figure 2) A = 9 B = 15 C = 21 D = 25 E = 31 F = 37	Termination length 01 = 109 02 = .140 03 = .172	Shell finish C = Cadmium N = Electroless nickel (space application)	Hardware N = No jackpost P = Jackpost attached

CONCLUDING MATERIAL

Custodians.
Army - CR
Navy - EC
Air Force - 85
NASA - NA
Review activities
Army - MI
Navy - AS, SH
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AT, ME
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent
DLA - ES

(Project 5935-3921-01)

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